

Task 1

3 Clients

Map Statistics (In-Memory Format: BINARY)

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Member ^	^ Entries	^ Gets	^ Puts	^ Removals	
127.0.0.1:5701	360	0	360	0	
127.0.0.1:5702	315	0	315	0	
127.0.0.1:5703	325	0	325	0	
TOTAL	1,000	0	1,000	0	

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2 Clients

Map Statistics (In-Memory Format: BINARY)

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Member ^	^ Entries	^ Gets	^ Puts	^ Removals	
127.0.0.1:5701	530	0	360	0	
127.0.0.1:5702	470	0	315	0	
TOTAL	1,000	0	675	0	

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1 Client

Map Statistics (In-Memory Format: BINARY)

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Member ^	^ Entries	^ Gets	^ Puts	^ Removals	
127.0.0.1:5701	530	0	360	0	
127.0.0.1:5702	470	0	315	0	
TOTAL	1,000	0	675	0	

1 – 2 of 2 Rows [10](#) [v](#)

Without locks:

```
(venv) nazar@nazar-HP-EliteBook-850-G5:~/3Year/hz_homework$ python3 main.py  
Final result (no locks): 10000
```

With pessimistic lock:

```
(venv) nazar@nazar-HP-EliteBook-850-G5:~/3Year/hz_homework$ python3 main.py  
Elapsed (pessimistic lock): 7.69  
Final result (pessimistic lock): 20000
```

With optimistic lock:

```
(venv) nazar@nazar-HP-EliteBook-850-G5:~/3Year/hz_homework$ python3 main.py  
Elapsed (optimistic lock): 3.57  
Final result (optimistic lock): 30000
```

Оптимістичний підхід працює швидше.

Consumer-Producer:

```
nazar@nazar-HP-EliteBook-850-G5:~/3Year/hz_homework$ python3 main.py
```

```
Produced: 1  
Consumer-A consumed: 1  
Produced: 2  
Consumer-B consumed: 2  
Produced: 3  
Consumer-A consumed: 3  
Consumer-B consumed: 4  
Produced: 4  
Consumer-A consumed: 5  
Produced: 5  
Consumer-B consumed: 6  
Produced: 6  
Consumer-A consumed: 7  
Produced: 7  
Produced: 8  
Consumer-B consumed: 8  
Consumer-A consumed: 9  
Produced: 9  
Produced: 10  
Consumer-B consumed: 10
```

Consumer-A consumed: 11
Produced: 11
Consumer-B consumed: 12
Produced: 12
Produced: 13
Consumer-A consumed: 13
Produced: 14
Consumer-B consumed: 14
Produced: 15
Consumer-A consumed: 15
Consumer-B consumed: 16
Produced: 16
Produced: 17
Consumer-A consumed: 17
Produced: 18
Consumer-B consumed: 18
Consumer-A consumed: 19
Produced: 19
Produced: 20
Consumer-B consumed: 20
Produced: 21
Consumer-A consumed: 21
Consumer-B consumed: 22
Produced: 22
Consumer-A consumed: 23
Produced: 23
Produced: 24
Consumer-B consumed: 24
Produced: 25
Consumer-A consumed: 25
Consumer-B consumed: 26
Produced: 26
Produced: 27
Consumer-A consumed: 27
Produced: 28
Consumer-B consumed: 28
Produced: 29
Consumer-A consumed: 29
Consumer-B consumed: 30
Produced: 30
Produced: 31
Consumer-A consumed: 31
Produced: 32
Consumer-B consumed: 32

Consumer-A consumed: 33
Produced: 33
Consumer-B consumed: 34
Produced: 34
Produced: 35
Consumer-A consumed: 35
Consumer-B consumed: 36
Produced: 36
Produced: 37
Consumer-A consumed: 37
Consumer-B consumed: 38
Produced: 38
Produced: 39
Consumer-A consumed: 39
Produced: 40
Consumer-B consumed: 40
Produced: 41
Consumer-A consumed: 41
Produced: 42
Consumer-B consumed: 42
Consumer-A consumed: 43
Produced: 43
Produced: 44
Consumer-B consumed: 44
Consumer-A consumed: 45
Produced: 45
Produced: 46
Consumer-B consumed: 46
Consumer-A consumed: 47
Produced: 47
Consumer-B consumed: 48
Produced: 48
Consumer-A consumed: 49
Produced: 49
Produced: 50
Consumer-B consumed: 50
Consumer-A consumed: 51
Produced: 51
Produced: 52
Consumer-B consumed: 52
Produced: 53
Consumer-A consumed: 53
Produced: 54
Consumer-B consumed: 54

Produced: 55
Consumer-A consumed: 55
Produced: 56
Consumer-B consumed: 56
Produced: 57
Consumer-A consumed: 57
Produced: 58
Consumer-B consumed: 58
Produced: 59
Consumer-A consumed: 59
Produced: 60
Consumer-B consumed: 60
Produced: 61
Consumer-A consumed: 61
Consumer-B consumed: 62
Produced: 62
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Consumer-A consumed: 63
Produced: 64
Consumer-B consumed: 64
Produced: 65
Consumer-A consumed: 65
Produced: 66
Consumer-B consumed: 66
Consumer-A consumed: 67
Produced: 67
Produced: 68
Consumer-B consumed: 68
Produced: 69
Consumer-A consumed: 69
Produced: 70
Consumer-B consumed: 70
Produced: 71
Consumer-A consumed: 71
Consumer-B consumed: 72
Produced: 72
Produced: 73
Consumer-A consumed: 73
Produced: 74
Consumer-B consumed: 74
Produced: 75
Consumer-A consumed: 75
Produced: 76
Consumer-B consumed: 76

Produced: 77
Consumer-A consumed: 77
Consumer-B consumed: 78
Produced: 78
Produced: 79
Consumer-A consumed: 79
Consumer-B consumed: 80
Produced: 80
Produced: 81
Consumer-A consumed: 81
Consumer-B consumed: 82
Produced: 82
Produced: 83
Consumer-A consumed: 83
Produced: 84
Consumer-B consumed: 84
Produced: 85
Consumer-A consumed: 85
Produced: 86
Consumer-B consumed: 86
Consumer-A consumed: 87
Produced: 87
Consumer-B consumed: 88
Produced: 88
Produced: 89
Consumer-A consumed: 89
Consumer-B consumed: 90
Produced: 90
Produced: 91
Consumer-A consumed: 91
Produced: 92
Consumer-B consumed: 92
Produced: 93
Consumer-A consumed: 93
Consumer-B consumed: 94
Produced: 94
Produced: 95
Consumer-A consumed: 95
Produced: 96
Consumer-B consumed: 96
Consumer-A consumed: 97
Produced: 97
Produced: 98
Consumer-B consumed: 98

Consumer-A consumed: 99
Produced: 99
Consumer-B consumed: 100
Produced: 100